

Micro I, class 12

I proved the first welfare theorem that a competitive equilibrium allocation is Pareto optimal by using the Kuhn-Tucker theorem from concave programming. The details are in my notes on the two-good quasilinear economy. I will begin Thursday by working through MWG 10.C.3, involving the ACM problem I concluded with. (This problem is mathematically the same as MWG 10.C.4, which does the same thing for the aggregate-utility-maximization problem.)

One of the reasons I like to close the first half of the term with MWG 10.C.3 is that it reviews many of the ideas we have used in the course so far. Thinking through it carefully might be an economical way to *begin* to organize your thoughts for the midterm.

One thing I did not mention is that the two-good quasilinear economy satisfies the Gorman conditions for a representative consumer. That consumer has preferences represented by $U(X, M) = \Phi(X) + M$, where $\Phi(\cdot)$ is the value function for problem (AUM) in my notes (the same thing as the function $\gamma(\cdot)$ in MWG 10.C.4). The economy also has a representative firm with aggregate cost function $C(Q)$ and aggregate production function $F(z) = C^{-1}(z)$. The representative-firm fact does not depend on what the technologies of the individual firms are and extends generally to competitive economies with arbitrary production sets (MWG, Prop 5.E.1(ii), reproduced in ASMT Va).

The last issue will be about *existence* of an equilibrium. In this case, existence will follow from an application of the elementary *intermediate value theorem* from calculus, a special kind of a fixed-point theorem. The discussion will be centered on the Lemma on page 3 of my notes on the two-good quasilinear economy, and the two paragraphs following it. I will keep my discussion of this point brief, so we can have a conversation the rest of the scheduled class that can spill over to my after-class office hour.